call past—were things more ordered? Why was the great pack of cards of the universe in order in the past? Why, in the past, was entropy lower?

If we observe a phenomenon that *begins* in a state of lower entropy, it is clear why entropy increases—because in the process of reshuffling, everything becomes disordered. But why do the phenomena that we observe around us in the cosmos *begin* in a state of lower entropy in the first place?

Here we get to the key point. If the first twenty-six cards in a pack are all red and the next twenty-six are all black, we say that the configuration of the cards is "particular," that it is "ordered." This order is lost when the pack is shuffled. The initial ordered configuration is a configuration "of low entropy." But notice that it is particular if we look at the *color* of the cards—red or black. It is particular because I am looking at the color. Another configuration will be particular if the first twenty-six cards consist of only hearts and spades. Or if they are all odd numbers, or the twenty-six most creased cards in the pack, or exactly the same twenty-six of three days ago. . . . Or if they share any other characteristic. If we think about it carefully, *every configuration is particular*, every configuration is singular, if we look at *all* of its details, since every configuration always has something about it that characterizes it in a unique way. Just as, to its mother, every child is particular and unique.

It follows that the notion of certain configurations being more particular than others (twenty-six red cards followed by twenty-six black, for example) makes sense only if I limit myself to noticing only certain aspects of the cards (in this case, the colors). If I distinguish between all the cards, the configurations are all equivalent: none of them is more or less particular than others.[18] The notion of "particularity" is born only at the moment we begin to see the universe in a blurred and approximate way.

Boltzmann has shown that entropy exists because we describe the world in a blurred fashion. He has demonstrated that entropy is precisely the quantity that counts *how many* are the different configurations that our blurred vision does *not* distinguish between. Heat, entropy, and the lower entropy of the past are notions that belong to an approximate, statistical description of nature.

The difference between past and future is deeply linked to this blurring. . . . So if I could take into account all the details of the exact, microscopic state of the world, would the characteristic aspects of the flowing of time disappear?

Yes. If I observe the microscopic state of things, then the difference between past and future vanishes. The future of the world, for instance, is determined by its present state—though neither more nor less than is the past.[19] We often say that causes precede effects and yet, in the elementary grammar of things, there is no distinction between "cause" and "effect."*